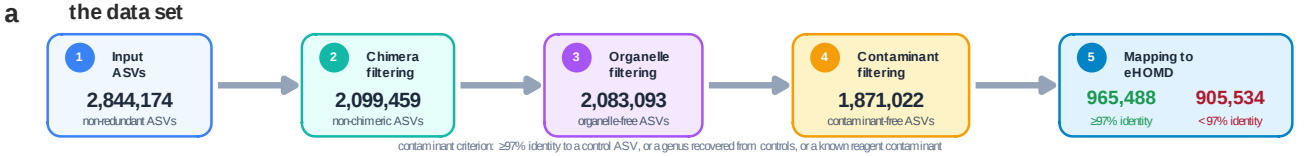
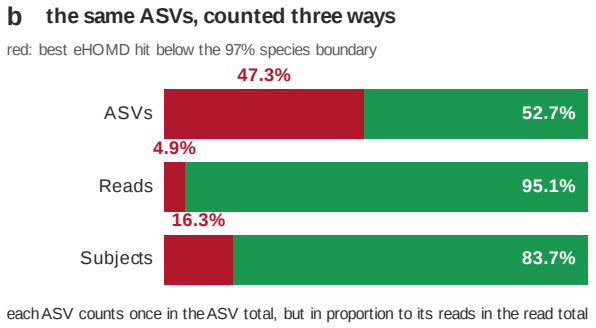
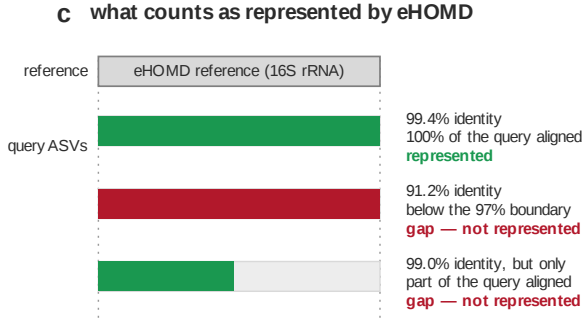
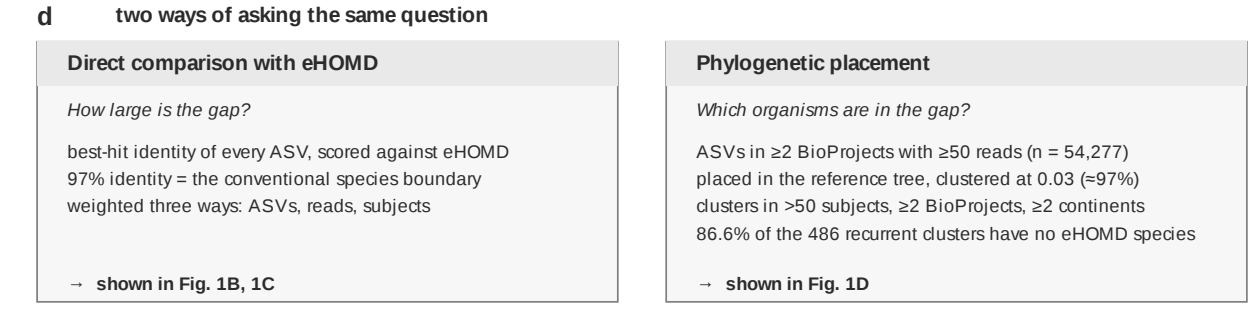

a
the data set
1
Input ASVs
2,844,174
non-redundant ASVs
2
Chimera filtering
2,099,459
non-chimeric ASVs
3
Organelle filtering
2,083,093
organelle-free ASVs
4
Contaminant filtering
1,871,022
contaminant-free ASVs
5
Mapping to eHOMD
965,488
≥97% identity
905,534
<97% identity
contaminant criterion: ≥97% identity to a control ASV, or a genus recovered from controls, or a known reagent contaminant


b
the same ASVs, counted three ways
red: best eHOMD hit below the 97% species boundary
47.3%
ASVs
52.7%
4.9%
Reads
95.1%
16.3%
Subjects
83.7%
each ASV counts once in the ASV total, but in proportion to its reads in the read total


c
what counts as represented by eHOMD
reference
eHOMD reference (16S rRNA)
query ASVs
99.4% identity
100% of the query aligned
represented
91.2% identity
below the 97% boundary
gap — not represented
99.0% identity, but only
part of the query aligned
gap — not represented


d
two ways of asking the same question
Direct comparison with eHOMD
How large is the gap?
best-hit identity of every ASV, scored against eHOMD
97% identity = the conventional species boundary
weighted three ways: ASVs, reads, subjects
→ shown in Fig. 1B, 1C
Phylogenetic placement
Which organisms are in the gap?
ASVs in ≥2 BioProjects with ≥50 reads (n = 54,277)
placed in the reference tree, clustered at 0.03 (≈97%)
clusters in >50 subjects, ≥2 BioProjects, ≥2 continents
86.6% of the 486 recurrent clusters have no eHOMD species
→ shown in Fig. 1D
both read the same ~97% species boundary